COMMUNITY BLOG

Open Margin database schema

MongoDB collections for the 3-week MVP. This is the shared design for auth, posts, the public site, and the AI writing assistant. Stack: Mongoose on the Express API.

Two collections already exist in the API: Tag and AiUsage. Tag still needs a slug. Everything else on this page is the schema the team should build. A visitor is not a database role. Visitors are people who are not logged in.

Roles stored on User: author (every registered account) and admin (one or more accounts). Passwords are hashed and never returned by the API.

Collections

Collection	What it is for
User	Accounts, profiles, and who is blocked
Category	Admin-managed sections such as City or Food
Tag	Reusable labels. The assistant prefers tags that already exist
Post	Drafts and published stories, including the accepted summary
Comment	Comments on a published post
Like	One like per person per post
Report	A post or comment flagged for an admin
AiUsage	How many assistant requests a person has used today

User

New collection

Sign up, log in, and the public author profile. Blocking a user is an admin action.

Field	Type	Rules
name	String	Required. Trimmed.
email	String	Required. Unique. Stored lowercase.
password	String	Required. Hashed. Never selected in API responses.
role	String	author or admin. Default author.
bio	String	Shown on the author profile. Optional.
avatar	Object	url and publicId from the image host. Optional.
isBlocked	Boolean	Default false. A blocked user cannot publish or engage.
createdAt	Date	Set by Mongoose timestamps.
updatedAt	Date	Set by Mongoose timestamps.

Indexes. Unique index on email.

Category

New collection. Managed by an admin.

The public site filters posts by category. The homepage cards use this name.

Field	Type	Rules
name	String	Required. Unique. Example: City, Food, Craft.
slug	String	Required. Unique. Used in the category URL.
createdAt	Date	Set by Mongoose timestamps.
updatedAt	Date	Set by Mongoose timestamps.

Indexes. Unique indexes on name and slug.

Tag

Already in the API. Add slug before the editor depends on it.

Suggest tags returns at most five tags and reuses an existing spelling when the name matches.

Field	Type	Rules
name	String	Required. Unique. Trimmed.
slug	String	Required. Unique. Used when filtering by tag.
createdAt	Date	Set by Mongoose timestamps.
updatedAt	Date	Set by Mongoose timestamps.

Indexes. Unique indexes on name and slug.

Post

New collection

An author saves a draft or publishes. The editor writes AI text into this document only after Accept.

Field	Type	Rules
author	ObjectId → User	Required.
title	String	Required. Accepting a suggested title writes here.
slug	String	Required. Unique. Used on the post page.
body	String	Required. The author's own draft. Improve this paragraph replaces only the selected text after Accept.
excerpt	String	The 2–3 sentence homepage summary. Filled when the author accepts it or types it.
coverImage	Object	url and publicId. Optional while the post is a draft.
category	ObjectId → Category	Required before publish.
tags	ObjectId[] → Tag	At most five.
status	String	draft or published. Default draft.
publishedAt	Date	Set when status becomes published.
createdAt	Date	Set by Mongoose timestamps.
updatedAt	Date	Set by Mongoose timestamps.

Indexes. { status, publishedAt }, { author, status }, { category, status }. Text index on title, excerpt, and body for search.

Comment

New collection

Logged-in users comment on a published post. An admin can remove a comment.

Field	Type	Rules
post	ObjectId → Post	Required.
author	ObjectId → User	Required.
body	String	Required.
createdAt	Date	Set by Mongoose timestamps.
updatedAt	Date	Set by Mongoose timestamps.

Indexes. { post, createdAt }.

Like

New collection

A person can like a post once. Liking again removes the row.

Field	Type	Rules
post	ObjectId → Post	Required.
user	ObjectId → User	Required.
createdAt	Date	Set when the like is created.

Indexes. Unique index on { user, post }.

Report

New collection

A logged-in user reports a post or a comment. The admin list shows open reports.

Field	Type	Rules
reporter	ObjectId → User	Required.
targetType	String	post or comment.
targetId	ObjectId	The reported post or comment.
reason	String	Required.
status	String	open or resolved. Default open.
resolvedBy	ObjectId → User	The admin who closed it. Empty while open.
createdAt	Date	Set by Mongoose timestamps.
updatedAt	Date	Set by Mongoose timestamps.

Indexes. { status, createdAt }.

AiUsage

Already in the API.

Counts assistant requests per person per day in Africa/Kigali. The limit is 20. Suggestions themselves are not stored.

Field	Type	Rules
userId	String	Required. Matches the user id on the token.
date	String	Required. YYYY-MM-DD in Africa/Kigali.
count	Number	Required. Default 0. Increased only after the request is valid.
createdAt	Date	Set by Mongoose timestamps.
updatedAt	Date	Set by Mongoose timestamps.

Indexes. Unique index on { userId, date }.

How the assistant writes into Post

The four editor buttons return JSON only. Nothing from the model is saved until the author clicks Accept, and they can edit the suggestion first. The assistant never writes a whole post.

Button	Returns	Accept writes
Suggest titles	3 titles	Author picks one. Accept sets Post.title.
Write a summary	2–3 sentences	Accept sets Post.excerpt.
Suggest tags	Up to 5 tags	Accept sets Post.tags, reusing existing Tag names.
Improve this paragraph	Clearer selected text	Accept replaces that span in Post.body.

Leave these out until the MVP works

- Follow: follower and following, unique on that pair, for a Following feed.
- viewCount on Post.
- A machine translation of a post between English and Kinyarwanda, stored on the post and marked as machine-translated.
- AI moderation adds a Report with source set to ai. It does not need its own collection.

Out of scope for this schema: the assistant writing a whole post, paid subscriptions, ads, and private messages.